

Project Implementation Plan

Intelligent Document Verification & Tamper Detection System

1. Project Overview

This project is an advanced security platform designed to verify the authenticity of educational and identity documents. It combines Computer Vision, Deep Learning, OCR, and Blockchain technology to detect even the most sophisticated digital and physical tampering.

2. Technical Architecture

The system follows a Decoupled Micro-services Architecture:

- Frontend: React.js (Vite)
- Backend: Flask (Python)
- Database: MongoDB
- Blockchain: Custom Python Ledger

3. Core Implementation Modules

Phase A: Image Preprocessing & Registration

- SIFT: Detects key points for alignment.
- Homography: Aligns user upload with ground truth.

Phase B: Visual Feature Extraction

- CNN (VGG16): Deep semantic features.
- HOG: Structural layout.
- LBP: Texture anomaly detection.
- dHash: Visual fingerprinting.

Phase C: OCR & Data Verification

- Affinda AI: High-precision extraction.
- EasyOCR: Fallback engine.

Phase D: Blockchain Integrity

- SHA-256 Chaining: Immutable record keeping.

4. Tamper Detection Logic

A document is flagged as TAMPERED if:

1. Metadata Check: Traces of Photoshop/GIMP found.
2. Field Mismatch: Marks/Name mismatch.
3. Visual Variance: Hash variance exceeds 15%.
4. Local Anomaly: Specific grid cell texture mismatch.

5. Technology Stack Summary

Languages: Python, JavaScript

Computer Vision: OpenCV, Scikit-Image

Deep Learning: TensorFlow (VGG16)

OCR: Affinda, EasyOCR

Database: MongoDB

Backend: Flask

Frontend: React, Framer Motion

6. Security Workflow

1. Admin Uploads: Original data hashed & stored.
2. User Uploads: Copy aligned & analyzed.
3. Decision: Tampered regions highlighted; authenticity confirmed via Blockchain.